

LHR-01-11.19

Roll No

(To be filled in by the candidate)

MATHEMATICS

Academic Sessions 2015 – 2017 to 2018 – 2020

PAPER – I (Objective Type)

219-(INTER PART – I)

Time Allowed : 30 Minutes

GROUP – I

Maximum Marks : 20

PAPER CODE = 6195

Note : Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

1-1	If $x - a$ is a factor of polynomial $f(x)$, then $f(a)$ is : (A) $= 0$ (B) < 0 (C) > 0 (D) $\neq 0$
2	If ${}^nC_5 = {}^nC_4$, then n is : (A) 9 (B) 7 (C) 6 (D) 5
3	The multiplicative inverse of $(1, -2) =$: (A) $(\frac{1}{5}, \frac{-2}{5})$ (B) $(\frac{-1}{5}, \frac{-2}{5})$ (C) $(\frac{1}{5}, \frac{2}{5})$ (D) $(\frac{-1}{5}, \frac{2}{5})$
4	9th term in the sequence $\frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \dots$ is : (A) $\frac{1}{13}$ (B) $\frac{1}{15}$ (C) $\frac{1}{17}$ (D) $\frac{1}{19}$
5	The contrapositive of $\sim p \rightarrow \sim q$ is : (A) $p \rightarrow q$ (B) $q \rightarrow p$ (C) $\sim q \rightarrow \sim p$ (D) $\sim q \rightarrow p$
6	From the identity $5x + 4 = A(x - 1) + B(x + 2)$, then value of $B =$: (A) -3 (B) 3 (C) -2 (D) 2
7	The sum of four 4 th roots of 16 is : (A) 0 (B) 2 (C) 4 (D) 16
8	If $\begin{bmatrix} x-3 & 1 \\ -5 & -4 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -5 & -4 \end{bmatrix}$, then $x =$: (A) 5 (B) -5 (C) -1 (D) 1
9	The arithmetic mean between $\sqrt{2}$ and $3\sqrt{2}$ is : (A) $3\sqrt{2}$ (B) $2\sqrt{2}$ (C) $4\sqrt{2}$ (D) $\sqrt{2}$
10	If $A = \begin{bmatrix} 1 & 2 & -2 \\ 0 & 0 & 5 \\ 6 & 7 & 3 \end{bmatrix}$, then $A_{33} =$: (A) -1 (B) 1 (C) 3 (D) 0
11	Period of $\cot \theta$ is : (A) π (B) 2π (C) $\frac{\pi}{2}$ (D) $\frac{3\pi}{2}$

(Turn Over)

1-12	Number of signals can be made with 4 flags when one flag is used at a time are : (A) 4C_0 (B) 4C_1 (C) 4C_2 (D) 4C_3
13	The equation $\sin^2 x - \sec x = \frac{3}{4}$ is called : (A) Trigonometric equation (B) Linear equation (C) Quadratic equation (D) Quantic equation
14	$3\sin \alpha - 4\sin^3 \alpha = :$ (A) $\sin \alpha$ (B) $\sin 2\alpha$ (C) $\sin 3\alpha$ (D) $\sin 4\alpha$
15	Domain of the function $y = \sin^{-1} x$ is : (A) $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ (B) $-1 \leq y \leq 1$ (C) $-1 \leq x \leq 1$ (D) $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
16	Francesco Mourollico devised the method of : (A) Partial fraction (B) Induction (C) Logarithms (D) Binomial
17	If $\ell = 35$ cm and $\theta = 1$ rad, then $r = :$ (A) 35° (B) 35 cm (C) 35 rad (D) 35 m
18	In any ΔABC with usual notations, $\frac{\Delta}{s-c} = :$ (A) r (B) r_1 (C) r_2 (D) r_3
19	The general term in the expansion of $(a+x)^n$ is : (A) $\binom{n}{a} a^{n-r} x^r$ (B) $\binom{n}{x} a^{n-r} x^r$ (C) $\binom{n}{r} a^{n-r} x^r$ (D) $\binom{n}{r} a^{n-r} x$
20	If sides of a ΔABC are $a = 4584$, $b = 5140$ and $c = 3624$, then greatest angle will be : (A) α (B) β (C) γ (D) a

Roll No

LHR-G1-11.19

(To be filled in by the candidate)

(Academic Sessions 2015 – 2017 to 2018 – 2020)

MATHEMATICS

219-(INTER PART – I)

Time Allowed : 2.30 hours

PAPER – I (Essay Type)

GROUP – I

Maximum Marks : 80

SECTION – I**2. Write short answers to any EIGHT (8) questions :****16**

- (i) If z_1 and z_2 are complex numbers then show that $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$
- (ii) Find out real and imaginary parts of $(\sqrt{3} + i)^3$
- (iii) Factorize $a^2 + 4b^2$
- (iv) Define power set of a set and give an example.
- (v) Define a bijective function.
- (vi) Construct truth table and show that the statement $\sim(p \rightarrow q) \rightarrow p$ is a tautology or not.
- (vii) Find the matrix X if $X \begin{bmatrix} 5 & 2 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 12 & 3 \end{bmatrix}$
- (viii) For the matrix $A = \begin{bmatrix} 1 & -2 & 3 \\ -2 & 3 & 1 \\ 4 & -3 & 2 \end{bmatrix}$ find cofactor A_{12}
- (ix) Without expansion show that $\begin{vmatrix} \alpha & \beta + \gamma & 1 \\ \beta & \gamma + \alpha & 1 \\ \gamma & \alpha + \beta & 1 \end{vmatrix} = 0$
- (x) When $x^4 + 2x^3 + kx^2 + 3$ is divided by $(x - 2)$, the remainder is 1. Find the value of k .
- (xi) If α, β are the roots of $ax^2 + bx + c = 0$, $a \neq 0$ then find the value of $\alpha^2 + \beta^2$
- (xii) The sum of a positive number and its square is 380. Find the number.

3. Write short answers to any EIGHT (8) questions :**16**

- (i) Define partial fraction.
- (ii) In the identity $7x + 25 = A(x + 4) + B(x + 3)$, calculate values of A and B .
- (iii) Resolve $\frac{1}{x^2 - 1}$ into partial fractions.
- (iv) Write the first four terms of the sequence, if $a_n - a_{n-1} = n + 2$, $a_1 = 2$
- (v) Which term of the arithmetic sequence 5, 2, -1, ---- is -85.
- (vi) Find three A.Ms between 3 and 11.
- (vii) If $\frac{1}{a}, \frac{1}{b}$ and $\frac{1}{c}$ are in G.P, show that common ratio is $\pm \sqrt{\frac{a}{c}}$
- (viii) Insert two G.Ms between 2 and 16.
- (ix) Find the value of n when ${}^nC_{10} = \frac{12 \times 11}{2!}$
- (x) Show that $\frac{n^3 + 2n}{3}$ represents an integer for $n = 2, 3$.
- (xi) Expand $\left(1 - \frac{3}{2}x\right)^{-2}$ upto 4 terms.
- (xii) If x is so small that its square and higher power can be neglected, then show that $\frac{\sqrt{1+2x}}{\sqrt{1-x}} \approx 1 + \frac{3}{2}x$

(Turn Over)

4. Write short answers to any NINE (9) questions :

12

- (i) Find ℓ , if $\theta = 65^\circ 20'$, $r = 18$ mm
- (ii) Prove $\sin^2 \frac{\pi}{6} : \sin^2 \frac{\pi}{4} : \sin^2 \frac{\pi}{3} : \sin^2 \frac{\pi}{2} = 1:2:3:4$
- (iii) Prove $\cos^2 \theta - \sin^2 \theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$
- (iv) Prove that $\tan 56^\circ = \frac{\cos 11^\circ + \sin 11^\circ}{\cos 11^\circ - \sin 11^\circ}$
- (v) Prove $\frac{1 - \cos \alpha}{\sin \alpha} = \tan \frac{\alpha}{2}$
- (vi) Prove $\cos 20^\circ + \cos 100^\circ + \cos 140^\circ = 0$
- (vii) Find the period of $\tan \frac{x}{7}$
- (viii) In $\triangle ABC$, $\beta = 60^\circ$, $\gamma = 15^\circ$, $b = \sqrt{6}$, find c .
- (ix) If $a = 200$, $b = 120$, $\gamma = 150^\circ$, find the area of a triangle ABC
- (x) Prove that $r_1 r_2 r_3 = rs^2$
- (xi) Prove $\sin(2 \cos^{-1} x) = 2x\sqrt{1-x^2}$
- (xii) Solve $1 + \cos x = 0$
- (xiii) Find the solutions of $\sin x = -\frac{\sqrt{3}}{2}$ in $[0, 2\pi]$

SECTION - II

Note : Attempt any THREE questions.

5. (a) Prove that all 2×2 non-singular matrices over the real field form a non-abelian group under multiplication. 5
- (b) Find three, consecutive numbers in G.P whose sum is 26 and their product is 216. 5
6. (a) Find the inverse of the matrix $A = \begin{bmatrix} 2 & 5 & -1 \\ 3 & 4 & 2 \\ 1 & 2 & -2 \end{bmatrix}$ by using row operation. 5
- (b) Prove that ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$ 5
7. (a) Solve the system of equations : 5

$$12x^2 - 25xy + 12y^2 = 0$$

$$4x^2 + 7y^2 = 148$$
- (b) If $y = \frac{1}{3} + \frac{1.3}{2!}(\frac{1}{3})^2 + \frac{1.3.5}{3!}(\frac{1}{3})^3 + \dots$ then prove that $y^2 + 2y - 2 = 0$ 5
8. (a) Prove that $\sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} = \sec \theta - \tan \theta$ where θ is not an odd multiple of $\frac{\pi}{2}$ 5
- (b) If α, β, γ are the angles of a triangle ABC, then show that : 5

$$\cot \frac{\alpha}{2} + \cot \frac{\beta}{2} + \cot \frac{\gamma}{2} = \cot \frac{\alpha}{2} \cot \frac{\beta}{2} \cot \frac{\gamma}{2}$$
9. (a) The sides of a triangle are $x^2 + x + 1$, $2x + 1$ and $x^2 - 1$. Prove that the greatest angle of the triangle is 120° . 5
- (b) Prove that $\tan^{-1} \frac{3}{4} + \tan^{-1} \frac{3}{5} - \tan^{-1} \frac{8}{19} = \frac{\pi}{4}$ 5

SECTION – I

2. Write short answers to any EIGHT (8) questions :

16

- (i) Prove the rule of addition $\frac{a}{c} + \frac{b}{c} = \frac{a+b}{c}$
- (ii) Find the multiplicative inverse of $(\sqrt{2}, -\sqrt{5})$
- (iii) Express the complex number $1+i\sqrt{3}$ in polar form.
- (iv) Write the power set of $\{a, \{b, c\}\}$
- (v) Show that the statement $p \rightarrow (p \vee q)$ is tautology.
- (vi) Prove that the identity element e in a group G is unique.
- (vii) If $A = \begin{bmatrix} 1 & -1 \\ a & b \end{bmatrix}$ and $A^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, find a and b
- (viii) If $B = \begin{bmatrix} 5 & -2 & 5 \\ 3 & -1 & 4 \\ -2 & 1 & -2 \end{bmatrix}$, find cofactor B_{21}
- (ix) If A is a skew-symmetric matrix, then show that A^2 is a symmetric matrix
- (x) Solve $x^{-2} - 10 = 3x^{-1}$.
- (xi) If α, β are the roots of $x^2 - px - p - c = 0$ then prove that $(1+\alpha)(1+\beta) = 1 - c$
- (xii) Discuss the nature of roots of the equation $x^2 - 5x + 6 = 0$

3. Write short answers to any EIGHT (8) questions :

16

- (i) Define proper fraction.
- (ii) If $\frac{x^2 - 10x + 13}{(x-1)(x^2 - 5x + 6)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$, find value of A
- (iii) If $\frac{x}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$, find value of B
- (iv) If the numbers $\frac{1}{k}, \frac{1}{2k+1}$ and $\frac{1}{4k-1}$ are in harmonic sequence, find k
- (v) Find sum of infinite geometric series $2 + 1 + 0.5 + \dots$
- (vi) Define geometric mean.
- (vii) If 5, 8 are two A.Ms between a and b , find a and b
- (viii) If $\frac{1}{a}, \frac{1}{b}$ and $\frac{1}{c}$ are in A.P, show that $b = \frac{2ac}{a+c}$
- (ix) Prove that ${}^nC_r = {}^nC_{n-r}$
- (x) Expand $(1+x)^{-1}$ upto 3 terms.
- (xi) Evaluate $\sqrt[3]{30}$ correct to three places of decimal.
- (xii) Check whether the statement $5^n - 2^n$ is divisible by 3 for $n = 2, 3$ is true or false.

(Turn Over)

4. Write short answers to any NINE (9) questions :

18

- (i) Find r , when $\ell = 56 \text{ cm}, \theta = 45^\circ$
- (ii) Find the values of all trigonometric functions for -15π
- (iii) Prove that $\frac{1 - \sin \theta}{\cos \theta} = \frac{\cos \theta}{1 + \sin \theta}$
- (iv) Express the difference $\cos 7\theta - \cos \theta$ as product.
- (v) Prove $\frac{1 - \cos \alpha}{\sin \alpha} = \tan \frac{\alpha}{2}$
- (vi) Find the value of $\cos 105^\circ$ without using calculator.
- (vii) Find the period of $3 \sin \frac{2x}{5}$
- (viii) With usual notations prove that $\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$
- (ix) Define in-circle of the triangle ABC.
- (x) State the law of tangent. (any two)
- (xi) Show that $\cos(2\sin^{-1} x) = 1 - 2x^2$
- (xii) Solve the equation for $\theta \in [0, \pi]$ $\cot^2 \theta = \frac{4}{3}$
- (xiii) Solve the equation for $\theta \in [0, \pi]$ $2\sin \theta + \cos^2 \theta - 1 = 0$

SECTION - II

Note : Attempt any THREE questions.

5. (a) If G is a group under the operation " $\ast$ " and $a, b \in G$, find the solutions of the equations : (i) $a \ast x = b$ (ii) $x \ast a = b$ 5
- (b) If 7th and 10th terms of an H.P are $\frac{1}{3}$ and $\frac{5}{21}$ respectively, find its 14th term 5
6. (a) Show that $\begin{vmatrix} a+\ell & a & a \\ a & a+\ell & a \\ a & a & a+\ell \end{vmatrix} = \ell^2(3a+1)$ 5
- (b) Prove that ${}^{n-1}C_r + {}^{n-1}C_{r-1} = {}^nC_r$ 5
7. (a) If α, β are the roots of $5x^2 - x - 2 = 0$ form the equation whose roots are $\frac{3}{\alpha}$ and $\frac{3}{\beta}$ 5
- (b) Use mathematical induction to prove that $n! > n^2$ for integral values of $n \geq 4$. 5
8. (a) A railway train is running on a circular track of radius 500 meters at the rate of 30 km per hour. Through what angle will it turn in 10 sec? 5
- (b) Reduce $\sin^4 \theta$ to an expression involving only function of multiples of θ raised to the first power. 5
9. (a) Prove that $r_1 r_2 + r_2 r_3 + r_3 r_1 = s^2$ 5
- (b) Prove that $\tan^{-1} A + \tan^{-1} B = \tan^{-1} \frac{A+B}{1-AB}$ 5

LHR-G2-11-19

Roll No _____ (To be filled in by the candidate)

MATHEMATICS - (Academic Sessions 2015 – 2017 to 2018 – 2020)

Q.PAPER – I (Objective Type) 219-(INTER PART – I)

Time Allowed : 30 Minutes

GROUP – II

Maximum Marks : 20

PAPER CODE = 6194

Note : Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

1-1	$\cos\left(\frac{3\pi}{2} - \theta\right)$ is equal to :	(A) $-\sin \theta$	(B) $\sin \theta$	(C) $\cos \theta$	(D) $-\cos \theta$
2	Probability of impossible event is :	(A) $\frac{1}{2}$	(B) 1	(C) 0	(D) 2
3	$2 \tan^{-1} A$ equals :	(A) $\tan^{-1}\left(\frac{A}{1-A^2}\right)$	(B) $\tan^{-1}\left(\frac{2A}{1-A^2}\right)$	(C) $\tan^{-1}\left(\frac{2A}{1+A^2}\right)$	(D) $\tan^{-1}\left(\frac{A}{1+A^2}\right)$
4	Which angle is quadrantal angle :	(A) 45°	(B) 60°	(C) 270°	(D) 120°
5	Solution of equation $\tan x = \frac{1}{\sqrt{2}}$ lies in the quadrants :	(A) I and II	(B) II and III	(C) I and III	(D) I and IV
6	Middle terms in the expansion of $(x+y)^{11}$ are :	(A) T_6, T_7	(B) T_5, T_6	(C) T_7, T_8	(D) T_8, T_9
7	If Δ is the area of a triangle ABC, then with usual notation $\Delta =$:	(A) $\frac{1}{2}bc \sin \beta$	(B) $\frac{1}{2}ab \sin \alpha$	(C) $\frac{1}{3}bc \sin \alpha$	(D) $\frac{1}{2}bc \sin \alpha$
8	Range of cotangent function is :	(A) N	(B) Z	(C) R	(D) C
9	Expansion of $(3-5x)^{\frac{1}{2}}$ is valid if :	(A) $ x < \frac{3}{5}$	(B) $ x < \frac{5}{3}$	(C) $ x < 5$	(D) $ x < 3$
10	With usual notation $R =$:	(A) $\frac{b}{2 \sin \gamma}$	(B) $\frac{a}{2 \sin \alpha}$	(C) $\frac{c}{2 \sin \alpha}$	(D) $\frac{a}{2 \sin \beta}$
11	The sum of the four fourth roots of 81 is :	(A) 0	(B) 81	(C) -81	(D) 3

(Turn Over)

HR-G2-11-19

(2)

1-12	The property $\forall a, b \in \mathbb{R}, a = b \Rightarrow b = a$ is called : (A) Commutative (B) Transitive (C) Symmetric (D) Reflexive
13	The value of $4! \cdot 0! \cdot 1!$ is : (A) 0 (B) 1 (C) 4 (D) 24
14	A square matrix $A = [a_{ij}]$ in which $a_{ij} = 0$ for all $i > j$ is called : (A) Upper triangular (B) Lower triangular (C) Symmetric (D) Skew-symmetric
15	$\sum_{k=1}^n (1)^k = :$ (A) $\frac{n(n-1)}{2}$ (B) $\frac{n}{2}$ (C) n (D) $\frac{n(n+1)}{2}$
16	If $b^2 - 4ac > 0$ but not a perfect square, then roots are : (A) Equal (B) Complex (C) Rational (D) Irrational
17	No term of geometric sequence can be : (A) 0 (B) 1 (C) 2 (D) 3
18	If A and B are two sets, then $A - B = :$ (A) $A \cup B^c$ (B) $A \cap B^c$ (C) $(A \cup B)^c$ (D) $(A \cap B)^c$
19	Partial fractions of $\frac{1}{x^3-1}$ will be of the form : (A) $\frac{A}{x+1} + \frac{Bx+C}{x^2+x+1}$ (B) $\frac{A}{x-1} + \frac{Bx+C}{x^2+x+1}$ (C) $\frac{A}{x-1} + \frac{Bx+C}{x^2-x+1}$ (D) $\frac{A}{x+1} + \frac{Bx+C}{x^2-x+1}$
20	If $A = [a_{ij}]_{2 \times 2}$, then $ kA = :$ (A) $ A $ (B) $k^2 A $ (C) $k A $ (D) $k A ^2$